

*Procedure #1***Prepared by:**  
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## CLAYTON BOILER START UP PROCEDURE

This procedure details the steps required to safely bring the Clayton Boiler online. It also includes provisions to bring the CCM plant online from a cold state.

### Clayton Boiler Walk Around and Auxiliary Equipment Checks

| Step | Operation                                                                                                                                                                    | Initial when Complete | Date and Time when Complete |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|
| 1    | Check the Plant Log Book to verify it is safe to start the Clayton Boiler.                                                                                                   |                       |                             |
| 2    | Ensure the <b>Red-Stop-Button</b> is in the off (pushed in) position.                                                                                                        |                       |                             |
| 3    | Check that the control power disconnect switch is switched on. The switch is located on the post in front of the boiler.                                                     |                       |                             |
| 4    | Walk around the Clayton Boiler, checking fittings, valve positions, gauges and gauge valves, interlock-and-permissive switches, linkages and connections, and the gas train. |                       |                             |
| 5    | Verify the Clayton Boiler's feed pump is ready for service and associated valves are in the proper position. Valves S3, S4. S21 Open. S22 Closed.                            |                       |                             |
| 6    | Check that there is ample feed water available (see the "Feedwater System Procedure" )                                                                                       |                       |                             |
| 7    | Check that the Clayton Boiler feedwater valves, including economizer valve S-21, are open to make sure feedwater will flow through the boiler.                               |                       |                             |

| 8                                                                                                                                                                                                                                                  | Open valve C-29, the separator steam trap outlet valve.                                                                                      |                       |                             |
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| 9                                                                                                                                                                                                                                                  | Check that the steam stop valve (chain operated valve) is closed.                                                                            |                       |                             |
| 10                                                                                                                                                                                                                                                 | Open the bypass around steam-stop-valve steam trap to help minimize water hammer. This steam trap is upstream of the backpressure regulator. |                       |                             |
| <b>Note: Do not make any adjustments to the backpressure regulator.</b>                                                                                                                                                                            |                                                                                                                                              |                       |                             |
| <b>Special Circumstance - Plant Cold Startup</b><br>This step is to be done when the plant piping is cold. Performing this step will reduce steam and hydraulic water hammer. <b>If the plant piping is hot, skip to "Clayton Boiler Startup".</b> |                                                                                                                                              |                       |                             |
| Step                                                                                                                                                                                                                                               | Operation                                                                                                                                    | Initial when Complete | Date and Time when Complete |
| 11                                                                                                                                                                                                                                                 | Check with the Plant Log Book to make sure there is no condition that will prevent the safe startup of the plant.                            |                       |                             |
| 12                                                                                                                                                                                                                                                 | Coordinate the start-up of the plant with the Building Operations Head, Mechanical Operations.                                               |                       |                             |
| 13                                                                                                                                                                                                                                                 | Partially open the drip leg drains on the main steam header and the low pressure header.                                                     |                       |                             |
| <b>Note:</b> Use drain hoses to direct cold condensate to the sewer drain to prevent flooding the floors. Stop using drain hoses when the condensate exceeds the temperature rating of the hose.                                                   |                                                                                                                                              |                       |                             |
| <b>Clayton Boiler Startup</b>                                                                                                                                                                                                                      |                                                                                                                                              |                       |                             |
| Step                                                                                                                                                                                                                                               | Operation                                                                                                                                    | Initial when Complete | Date and Time when Complete |
| 14                                                                                                                                                                                                                                                 | Pull out the Red-Stop-Button to initialize the Siemens PLC and the Clayton OIU. Wait 10 to 30 seconds for the units to initialize.           |                       |                             |

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| 15 | Press the Green-Start-Button. Check that the Clayton OIU entered “Fill Mode” and the feedwater pump has started.                                                                                                                        |  |  |
| 16 | Place the Clayton OIU into manual mode. This is done by pressing the Auto   Manual button on the CLAYTON OIU.                                                                                                                           |  |  |
| 17 | Set the manual fill rate to 25 percent.                                                                                                                                                                                                 |  |  |
| 18 | Check the feedwater pump is operating. Check the motor and pulley operation.                                                                                                                                                            |  |  |
| 19 | Perform a feedwater pump dead-head check by closing in on valve S-21 until the feedwater pump discharge pressure reads 100 PSig then fully open valve S-21. This is to be done once at initial startup.                                 |  |  |
| 20 | Partially open and leave partially open valve S-87 and check that hot feedwater is flowing. This proves feedwater is running through the boiler.                                                                                        |  |  |
| 21 | <p>Partially open valve C-31 to allow condensate to flow back to the Hotwell</p> <hr/> <p><b>Note:</b> This is necessary because when the boiler is not under pressure the trap bypass valve (C-31) allows flow through the boiler.</p> |  |  |
| 22 | Start the ignition sequence on the Clayton Boiler. This is done by selecting run on the Clayton OIU by pressing the Fill   Run button.                                                                                                  |  |  |
| 23 | Walk around the boiler during the purge sequence, looking for leaks and excessive vibration.                                                                                                                                            |  |  |
| 24 | Monitor the boiler for successful ignition of the main flame. Check for a flame at the burner.                                                                                                                                          |  |  |

**Note:** A stable flame signal must be achieved within the pilot trial-for-ignition (PTFI) period and maintained during the main flame trial-for-ignition (MFFI) period; if not, the ESC will lock out, requiring a manual reset

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| 25 | Slowly close in on valve C-31 as the steam pressure rises to prevent excess steam and condensate flow to the hotwell.                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |
| 26 | Verify the separator steam trap is working. The trap pressure gauge will be rising and falling, indicating the passing of condensate.                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |
| 27 | Close valve C-31 when the boiler pressure reaches 50 PSig.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  |
| 28 | Slowly open the main stop valve (chain operated) in controlled increments after the boiler pressure reads 50 PSig.                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |
| 29 | Verify that the separator temperature matches the saturated steam temperature at the pressure indicated on the boiler pressure gauges. A table outlining these values is printed on the Clayton Boiler control panel.                                                                                                                                                                                                                                                                                                                                           |  |  |
| 30 | <p>Continue slowly opening the main stop valve in controlled increments. Opening the valve in increments prevents the boiler's pressure from dropping too quickly and allows downstream piping to slowly warm up.</p> <hr/> <p><b>Note:</b> If starting the plant from a cold state, watch for condensate and then steam to flow from the free-blow floor drains on the headers.</p> <p>Close the free-blow floor drains once all condensate is drained and pressure is sufficient to for the traps to function and blow condensate through to the hotwell.</p> |  |  |
| 31 | If starting the plant from cold, monitor the steam pressure as it increases in all the steam headers.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |

**Note:** This process of warming the boiler and steam systems to full pressure and temperature from cold is expected to take between half-an-hour in warm weather and up to two hours during cold winter weather.

|                   |                                                                                                                                                                                                                                                         |  |  |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| 32                | Verify the steam trap between the separator outlet and the main stop valve is hot; to indicate it is passing hot condensate to the condensate return tank.                                                                                              |  |  |
| 33                | Observe the trap pressure gauge on the separator for proper trap timing as set out on page 2-4 of the Operations Manual.                                                                                                                                |  |  |
| 34                | Gradually increase the firing rate if the steam pressure stops rising or begins to fall.                                                                                                                                                                |  |  |
| 35                | Monitor the steam system as the steam pressure rises, ensure condensate is being drained and no water hammer is occurring.                                                                                                                              |  |  |
| 36                | Fully open the main stop valve when the boiler pressure reaches set-point.                                                                                                                                                                              |  |  |
| 37                | Maintain the boiler pressure close to set-point by gradually increasing the firing rate.                                                                                                                                                                |  |  |
| 38                | Place the Clayton Boiler into automatic when the boiler pressure matches set-point pressure. This is done by pressing the <b>Auto   Manual</b> button on the Clayton OIU.<br><br>Ensure the boiler is placed into remote mode before leaving the plant. |  |  |
| <b>Set-Points</b> |                                                                                                                                                                                                                                                         |  |  |

|                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| 39                                                               | <p>Verify the following set points on the Clayton OIU on a weekly basis:</p> <ul style="list-style-type: none"><li>• Modulating set-point: 90 PSig, as of July 2016;</li><li>• Cut-out set-point: 130PSig as of July 2016;</li><li>• Cut-in: 120 PSig as of July 2016.</li></ul> <hr/> <p><b>Note:</b> A CLAYTON OIU password is required to view and adjust set points. See Clayton Industries Instruction Manual for change details and operations.</p> |  |  |
| 40                                                               | Make an appropriate log book entry.                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |
| <hr/> <p style="text-align: center;"><b>End of Procedure</b></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |